Superstore Power BI Dashboard: Build Guide

1. Load the data (Power Query)

Get Data > Text/CSV and load all 5 files from the star_schema folder: Fact_Orders, Dim_Customer, Dim_Product, Dim_Date, Dim_Geography. They are already cleaned (stray tables removed, duplicate dropped, dates parsed). In Power Query, check types: Order Date / Ship Date / Date = **Date**, Sales / Profit / Discount = **Decimal**, Quantity = **Whole number**, keys = Text (Geo Key = Whole number). Then **Close & Apply**.

Note: Region and State need a home, so I added a small Dim_Geography next to the 4 dimensions you listed. 32 Product IDs have more than one name in the source; Dim_Product keeps the first one so the key stays unique.

2. Model (Model view)

From (one side)	To (many side)	Notes
Dim_Date[Date]	Fact_Orders[Order Date]	Active
Dim_Date[Date]	Fact_Orders[Ship Date]	Inactive (optional)
Dim_Customer[Customer ID]	Fact_Orders[Customer ID]	1:*, single direction
Dim_Product[Product ID]	Fact_Orders[Product ID]	1:*, single direction
Dim_Geography[Geo Key]	Fact_Orders[Geo Key]	1:*, single direction

Right-click Dim_Date > **Mark as date table** (column: Date). Hide the key columns in the Fact table. In Dim_Date, sort Month Name by Month Number.

3. DAX measures (create in a Measures table)

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Total Sales = SUM ( Fact_Orders[Sales] )
Total Profit = SUM ( Fact_Orders[Profit] )
Profit Margin % = DIVIDE ( [Total Profit], [Total Sales], 0 )
Total Orders = DISTINCTCOUNT ( Fact_Orders[Order ID] )
Total Quantity = SUM ( Fact_Orders[Quantity] )
Avg Discount % = AVERAGE ( Fact_Orders[Discount] )

Sales PY = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( Dim_Date[Date] ) )
YoY Sales Growth % = DIVIDE ( [Total Sales] - [Sales PY], [Sales PY] )

Profit PY = CALCULATE ( [Total Profit], SAMEPERIODLASTYEAR ( Dim_Date[Date] ) )
YoY Profit Growth % = DIVIDE ( [Total Profit] - [Profit PY], [Profit PY] )

Loss-Making Lines = CALCULATE ( COUNTROWS ( Fact_Orders ), Fact_Orders[Profit] < 0 )
Return Rate % =
DIVIDE (
    CALCULATE ( DISTINCTCOUNT ( Fact_Orders[Order ID] ), Fact_Orders[Returned] = "Yes" ),
    [Total Orders], 0 )
```

Set formats: Sales/Profit = Currency, Margin/Growth = Percentage (1 decimal). Growth is blank for 2015 (no prior year), which is expected.

4. Report page layout

Visual	Setup
KPI cards (x4)	Total Sales, Total Profit, Profit Margin %, Total Orders. Optional: add YoY Sales Growth % as the card sub-label.
Date slicer	Dim_Date[Date], style Between (slider). Add a Year slicer as tiles.
Region slicer	Dim_Geography[Region], style Tile/Buttons. Add Segment as a dropdown.
Monthly sales	Line chart: X = Dim_Date[Year-Month], Y = Total Sales. Add a second Y line with Sales PY, or use a Waterfall for month-on-month change.
Region treemap	Treemap: Group = Region, then State (drill down), Values = Total Sales, Tooltip = Profit Margin %.

Category treemap	Treemap: Group = Category > Sub-Category, Values = Total Sales. Conditional-format colour by Profit Margin %.
Profitability matrix	Matrix: Rows = Category > Sub-Category, Columns = Region, Values = Total Profit. Conditional formatting: red (negative) to green. A scatter of Sales vs Profit Margin % by Sub-Category works well beside it.

5. Numbers to sanity-check against

With no filters, your cards should show: **Sales $2,296,919** | **Profit $286,409** | **Margin 12.5%** | **Orders 5,009**. If they differ, a relationship or data type is off.

6. Export

.pbix: File > Save as. **PDF:** File > Export > Export to PDF (exports each report page). For the sharpest result, set the page size to 16:9 (1280x720 or larger) under Format > Page, hide the filter pane, and check the PDF after export. Keep the report to one clean page, plus an optional second page for the profitability matrix.